

Results Report: WT vs pifq single-cell RNA-seq pipeline

Executed results — real figures from a fresh run using the bibliography annotation (celltype);
code shown as in the methods paper

SingleCell Pipeline

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Abstract

A stepwise framework for inferring gene regulatory networks and pseudotime-based trajectories from single-cell RNA-sequencing data

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Plant development is a dynamic process in which new organs are continuously formed throughout an individual's lifetime. This continuous production of new organs enables plants to increase in size and generate tissues with diverse physiological and functional capabilities. Single-cell RNA sequencing (scRNA-seq) has emerged as a powerful approach for characterizing cell-type-specific transcriptomes at high resolution across tissues and species. However, the bioinformatics analyses required to extract biological information from scRNA-seq datasets can be challenging for researchers without formal training in bioinformatics. Here, we provide a step-by-step workflow for inferring gene regulatory networks and pseudotime-based developmental trajectories from scRNA-seq data. To facilitate the implementation and reproducibility of these analyses, we provide a Docker image and a GitHub repository (github.com/laboratoriorgit-lang/SingleCell) containing all scripts, helper functions, reference files, container configuration, and environment YAML files. The workflow was applied to publicly available scRNA-seq data from *Arabidopsis thaliana* seedlings and enables the identification of differentially expressed genes (DEGs) across cell types, gene regulatory networks, and developmental trajectories associated with epidermal differentiation. Although this guide focuses on scRNA-seq analysis in *A. thaliana*, the workflow can be adapted to other organisms using appropriate reference genomes, genome annotations, and experimental designs.

Introduction

Plant organs are not laid down once during embryogenesis but are produced continuously across the plant's life, from meristems that generate new roots, leaves and flowers on demand. This ongoing organogenesis depends on precisely coordinated, cell-type-specific gene expression programs, as neighboring cells within the same tissue can follow markedly different transcriptional trajectories depending on their developmental origin, position, and stage of differentiation. Bulk RNA-seq, which averages expression across every cell in a sample, is poorly suited to resolving this heterogeneity: rare cell types are diluted out, and transitions between cell states are collapsed into a single average that may not represent any individual cell.

Single-cell RNA sequencing (scRNA-seq) addresses this limitation directly by profiling transcriptomes one cell at a time, making it possible to resolve cell-type identity, cell-state transitions, and developmental trajectories within a tissue at a resolution bulk approaches cannot reach. Once cells are clustered and annotated, the same data can support several complementary analyses: differential expression between cell types or genotypes, inference of gene regulatory networks from co-expression patterns, and reconstruction of pseudotime trajectories that order cells along a developmental or differentiation axis. In practice, however, each of these steps involves a distinct set of tools, file formats and statistical choices, and combining them into a coherent, reproducible pipeline requires computational expertise that is not always available in plant biology labs, which can be a real barrier to adopting scRNA-seq despite its potential.

This guide provides a complete, reproducible workflow that carries scRNA-seq data from raw sequencing reads through all of these downstream analyses. Starting from FASTQ files, Cell Ranger is used to generate per-cell count matrices (Part 1). These matrices are then loaded into R, where cells are filtered, doublets are removed, samples are merged and batch-corrected with Harmony, and clusters are annotated to cell types using literature-curated marker genes (Part 2). From the annotated object, the pipeline builds pseudobulk profiles per cell type to identify differentially expressed genes between genotypes, runs GO enrichment on those gene sets, and infers a transcription-factor co-expression network to highlight candidate regulators (Part 3). Finally, the workflow switches to Python (scanpy, Palantir, scFates) to reconstruct pseudotime trajectories within a selected cell population, order cells along that trajectory, and identify genes whose expression trends define it (Part 4). Every step is packaged as a single container image, so the same software environment — R, Python, and every package version — is available to anyone who pulls the image, removing a major source of “works on my machine” irreproducibility.

To illustrate the complete workflow end to end, this document applies it to a public scRNA-seq dataset comparing wild-type and *pifq* mutant *Arabidopsis thaliana* seedlings, from raw Cell Ranger processing through cell-type annotation, differential expression, regulatory network inference, and a pseudotime trajectory of epidermal differentiation. Although the example organism is *A. thaliana*, none of the workflow logic is specific to it: the same steps apply to other species by substituting the reference genome, annotation, and marker gene set used for cell-type calling.

About this document

This methods-paper R Markdown documents the full WT vs *pifq* workflow: two samples, WT and *pifq*, processed and integrated together, then carried through three chapters — single-cell preprocessing, clustering and annotation (Part 2); pseudobulk differential expression, GO and the co-expression network (Part 3); and pseudotime trajectory analysis (Part 4). The code blocks are shown exactly as in the methods paper (not executed inline); the figures below are the **real outputs** of a fresh run of `workflow/step1_singlecell.R`, `workflow/step2_degs.R` and `workflow/step3_pseudotime.py`, written to a separate `results_rerun/` directory using the bibliography annotation (`celltype`).

The active annotation path is bibliography-based (`biblio_marks.txt -> celltype_curated`). Reference-transfer annotation is not part of this draft.

Minimum computing requirements

The table below summarizes the computing requirements for running the full pipeline on a personal computer, from raw FASTQ through the downstream Seurat/R and Python workflow. These are not hard limits; below them the pipeline still runs, just slower, and heavier steps (Cell Ranger, merging, PCA/UMAP) risk stalling or exhausting memory.

Component	Minimum	Recommended
CPU cores	4	16 or more
RAM	32 GB	64-128 GB for merged multi-sample objects
Disk space	80 GB free	100 GB free
Container runtime	Any OCI-compatible runtime (e.g. Docker)	Docker or Singularity/Apptainer

This workflow is packaged as a single public image (`psblab/scrnaseq:latest`, on Docker Hub); the `docker-compose` YAML used to build and run it is kept in the repository. Any container runtime able to run that image works; the setup below covers both Docker and Singularity/Apptainer (the latter common on HPC clusters, where Docker is usually not permitted).

Small pilot datasets can be run on a laptop or desktop if memory is sufficient. For full experiments, repeated clustering sweeps, Cell Ranger processing from FASTQ, or large merged Seurat objects, a server or external cluster is strongly recommended. Cell Ranger is CPU- and memory-intensive and is not installed inside the downstream `psblab/scrnaseq:latest` Docker image.

Recommended: persistent terminal sessions with `tmux` or `screen`

Several steps in this pipeline run for a long time without any interaction — Cell Ranger takes 30-60 minutes per sample, and the R/Python clustering and pseudotime steps can run for hours on large merged objects. If these are launched directly in an SSH session and the connection drops (network hiccup, laptop sleep, closed terminal), the process is killed along with it.

Running everything inside a `tmux` or `screen` session avoids this: the session keeps running on the server after you disconnect, and can be reattached later — from the same machine or a different one — to check progress or pick up where things were left off. This is strongly recommended any time this pipeline runs on a remote server or HPC cluster, and is still useful on a local machine to keep a session alive while doing other work.

Either tool works; `tmux` is used in the examples below.

Bash

```
tmux new -s scrnaseq      # start a new named session
# ... run Cell Ranger, Docker/Singularity, R, etc. inside it ...
# detach without stopping anything: Ctrl-b, then d

tmux attach -t scrnaseq   # reattach later
tmux ls                  # list running sessions
```

This pipeline’s R/Python environment ships as a single public container image (`psblab/scrnaseq:latest`, Docker or Singularity/Apptainer). It mounts the cloned repository as `/workspace`, so results are written back into the repo. Always pull the image rather than rebuilding it from the `Dockerfile`, so everyone gets the exact tested package versions (details and commands in “Container startup”, Part 1).

Once inside a container (Docker or Singularity) shell, stay inside it for the rest of the work. The R and Python packages this pipeline needs (Seurat, scanpy, Palantir, scFates, etc.) are installed only

inside the `psblab/scrnaseq:latest` image — they are not available on the bare host. Leaving the container shell (e.g. `exit`, or the shell dying) means the next command expecting those packages will fail. Pair this with `tmux/screen`: open a session, enter the container from inside it, and do the rest of the R/Python work (Parts 2-4) without leaving either until finished.

Installation instructions for `tmux`, `screen`, `Docker` and `Singularity/Apptainer` are in the Supplementary section at the end of this document.

1 Part 1 - Environment and raw data

1.1 Clone the repository

Choose or create a working directory, then clone the pipeline repository into it:

Bash

```
mkdir -p ~/my-analysis
cd ~/my-analysis
git clone https://github.com/laboratoriorgit-lang/SingleCell.git
cd SingleCell
```

All paths referenced later in this document (`workflow/`) are relative to this cloned repository. Its layout is:

```
SingleCell/
|-- Dockerfile
|-- docker-compose.yml
|-- README.md
|-- data/
|   |-- AtTFDB_loci.txt
|   |-- biblio_marks.txt
|   |-- Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta.gz
|   `-- Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf.gz
`-- workflow/
    |-- load_libraries.R                # R packages
    |-- load_libraries_python.py        # Python packages
    |-- ScRNA_Analysis_Functions.R      # helper functions (Steps 1-2)
    |-- ScRNA_Pseudotime_Functions.py   # helper functions (Step 3)
    |-- step0_cellranger.sh             # Part 1 - Cell Ranger (bash)
    |-- step1_singlecell.R              # Part 2 - single-cell analysis
    |-- step2_degs.R                    # Part 3 - pseudobulk DE + GO
    `-- step3_pseudotime.py             # Part 4 - pseudotime (Python)
```

The bibliography marker table (`data/biblio_marks.txt`) is part of the repository for this WT vs pifq run (see Section 1).

1.2 Container startup (Docker or Singularity)

The container mounts the cloned repository itself as `/workspace`, so results are written back into the repository (`results/`). Since results are not pushed upstream, this is not a problem. The same public image runs under either runtime; always pull it rather than rebuilding from the `Dockerfile`, so every user gets the exact tested package versions.

Docker. Mount the cloned repository and open a shell:

Bash

```
docker run -it --rm \
  -v $(pwd):/workspace \
  psblab/scrnaseq:latest /bin/bash
```

Or, from inside the cloned repository, `docker compose run r` uses the bundled `docker-compose.yml` (same image, same `/workspace` mount). On the current server the Docker container is already running and the project is mounted at `/workspace`.

Singularity / Apptainer. No root is required. Convert the public Docker Hub image to a local `.sif` once, then open a shell binding the repository to `/workspace`:

Bash

```
singularity build scrnaseq.sif docker://psblab/scrnaseq:latest
singularity exec --bind $(pwd):/workspace --pwd /workspace scrnaseq.sif /bin/bash
```

Apptainer imports the image environment, so the bundled Python (`/opt/venv`) and R stay on `PATH`, and it runs as the invoking user, so files under `results/` are owned by that user rather than by root. The `--pwd /workspace` flag starts the shell inside the bound repository (otherwise Apptainer warns that the host working directory does not exist in the container and falls back to `$HOME`).

Stay inside this shell for Parts 2-4. All R and Python packages this pipeline needs are installed only inside this container image; running them outside of it fails. Start this shell inside a `tmux/screen` session (see above) so it survives a dropped connection.

1.3 Raw data download and Cell Ranger processing

Run this whole section outside the Docker container, on a host, server, or HPC where Cell Ranger is installed. Cell Ranger is *not* bundled inside the `psblab/scrnaseq:latest` image (that image only covers the downstream R / Python workflow), so running `cellranger` inside the container returns **command not found**. Write the Cell Ranger outputs into the cloned repository directory so they are visible from inside the container (which mounts that same directory as `/workspace`).

Install Cell Ranger 9.0.1 (proprietary 10x Genomics software; this workflow uses version 9.0.1). Get it from the 10x Genomics *previous versions* page: <https://www.10xgenomics.com/support/software/cell-ranger/downloads/previous-versions>

1. Log in and accept the license.
2. Find **Cell Ranger 9.0.1** and copy its `curl` command.
3. That URL is signed and time-limited (it carries an `Expires=` timestamp tied to your account), so paste the current one — it cannot be hard-coded here or reused later. A plain `git clone` of the source repo does *not* give a ready-to-run binary.

Then extract it and add it to your `PATH`:

Bash

```
# Paste the signed curl command for 9.0.1 from the 10x page, e.g.:
curl -o cellranger-9.0.1.tar.gz "https://cf.10xgenomics.com/releases/cell-exp/\
cellranger-9.0.1.tar.gz?Expires=...&Key-Pair-Id=...&Signature=..."

tar -xzf cellranger-9.0.1.tar.gz
export PATH=$PWD/cellranger-9.0.1/bin:$PATH
cellranger --version
```

The single-cell workflow starts from Cell Ranger output, specifically the `outs/filtered_feature_bc_matrix/` folder for each sample. If only raw FASTQ files are available, first download or stage the FASTQs, then run Cell Ranger for each sample before starting the R workflow.

The WT vs pifq FASTQ files used in this run are public data from Han et al., 2023, *Nature Plants*, deposited at the CNCB Genome Sequence Archive under BioProject PRJCA016521, run accession CRA010863:

Sample	Run accession	Files
ScWT	CRR775298	CRR775298_f1.fastq.gz, CRR775298_r2.fastq.gz
Scpifq	CRR775297	CRR775297_f1.fastq.gz, CRR775297_r2.fastq.gz

Download the reads, then rename/symlink them to the naming Cell Ranger expects (<sample>_S1_L001_R1/R2_001.fastq.gz)

Bash

```
mkdir -p /workspace/fastq/CRA010863

wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775298/CRR775298_f1.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775298/CRR775298_r2.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775297/CRR775297_f1.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775297/CRR775297_r2.fastq.gz

cd /workspace/fastq/CRA010863
ln -s CRR775298_f1.fastq.gz ScWT_S1_L001_R1_001.fastq.gz
ln -s CRR775298_r2.fastq.gz ScWT_S1_L001_R2_001.fastq.gz
ln -s CRR775297_f1.fastq.gz Scpifq_S1_L001_R1_001.fastq.gz
ln -s CRR775297_r2.fastq.gz Scpifq_S1_L001_R2_001.fastq.gz
```

After FASTQs are available, run Cell Ranger for each sample. Cell Ranger itself is not redistributed in this repository; download it from the official 10x Genomics GitHub release (<https://github.com/10XGenomics/cellranger/releases/tag/cellranger-9.0.1>) and add it to your PATH. Run all commands in this section from the cloned repository directory, on a host/server/HPC where Cell Ranger is installed (not inside the container).

The reference genome (TAIR10.1) and Araport11 annotation ship compressed in the repository under `data/`, so no download is needed. Decompress them and build the Cell Ranger reference:

Bash

```
gunzip -k data/Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta.gz
gunzip -k data/Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf.gz

cellranger mkref \
  --genome=Ara \
  --fasta=data/Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta \
  --genes=data/Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf
```

Run Cell Ranger counts for each sample:

Bash

```
cellranger count --localcores=10 --id=ScWT --fastqs=/workspace/fastq/CRA010863 \  
  --sample=ScWT --transcriptome=Ara --create-bam false  
  
cellranger count --localcores=10 --id=Scpifq --fastqs=/workspace/fastq/CRA010863 \  
  --sample=Scpifq --transcriptome=Ara --create-bam false
```

--localcores=10 is used here as a portable default that keeps the example consistent across this report and the accompanying workflow scripts; it is not a hard requirement. Cell Ranger will happily use more or fewer cores, so raise or lower this value to match the number of CPU cores actually available on the machine running the pipeline (see the Recommended column above for guidance).

Use the resulting `filtered_feature_bc_matrix/` path in the sample manifest in Section 1.

2 Part 2 - Single-cell analysis in R

From this point on, the pipeline runs in R inside the Docker container. Start R:

Bash

R

The next two steps, Configuration and Initialization, set up the paths and helper functions used by every section that follows.

2.1 Configuration

This block defines the main paths used by the pipeline: the working directory, the folder where the workflow scripts are stored, and the folder where all results will be written. Edit these values before running the analysis if your project is mounted in a different location.

R

```
PIPELINE_DIR <- "/workspace/workflow"
DATA_DIR <- "/workspace/."
base_dir <- file.path(DATA_DIR, "results")
```

2.2 Initialization

This block prepares the R session before running the analysis. It loads the pipeline helper scripts, sets a fixed random seed for reproducibility, configures Seurat compatibility options, moves R into the project working directory, and creates the output folders used by the rest of the workflow. The seed is important because some analysis steps include random components; keeping it fixed helps obtain comparable results when the pipeline is rerun.

R

```
source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))
set.seed(1807)
options(Seurat.allow.s4 = FALSE)
setwd(DATA_DIR)
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)
output_dir <- base_dir
```

After this block runs, the R session is ready to execute the analysis sections. The helper functions are available, the project paths are fixed, and the results folders have been created. Each following section only needs to set `output_dir` to the corresponding folder before saving figures or objects.

2.3 Section 1 - Data loading and pre-filter QC

This section loads each sample from the Cell Ranger output and creates the first quality-control plots before filtering.

To use this block, edit the sample list with one entry per sample:

- **file:** path to the `filtered_feature_bc_matrix` folder, relative to `DATA_DIR`.
- **label:** short sample name used in plots and object labels.
- **condition:** experimental group or treatment condition.

Then edit `colors`, assigning one color to each sample `label`. These colors are used in the QC plots and later visualizations.

The mitochondrial and chloroplast gene patterns tell the pipeline how to identify organellar reads. For Arabidopsis, mitochondrial genes usually start with `ATMG` and chloroplast genes usually start with `ATCG`; those are the defaults used in this report.

After running the block, the pipeline creates one raw Seurat object per sample and saves the pre-filter QC plot in `results/01_qc/qc_prefilter.pdf`. This plot is used to inspect sample quality before choosing filtering thresholds in the next section.

R

```
output_dir <- dir_01
samples <- list(
  list(
    file      = "ScWT/outs/filtered_feature_bc_matrix",
    label     = "WT",
    condition = "WT"
  ),
  list(
    file      = "Scpifq/outs/filtered_feature_bc_matrix",
    label     = "pifq",
    condition = "pifq"
  )
)
colors <- c(
  "WT"      = "#66c2a5",
  "pifq"    = "#fc8d62"
)
mt_pattern <- "^ATMG"
cp_pattern <- "^ATCG"
seurat_list_raw <- load_seurat_samples(samples = samples,
                                       DATA_DIR = DATA_DIR,
                                       mt_pattern = mt_pattern,
                                       cp_pattern = cp_pattern)
plot_qc_batch(seurat_list_raw, colors, "qc_prefilter.pdf")
message("\nOK SECTION 1 COMPLETE: QC pre-filter plots saved")
```

Representative output from Section 1:

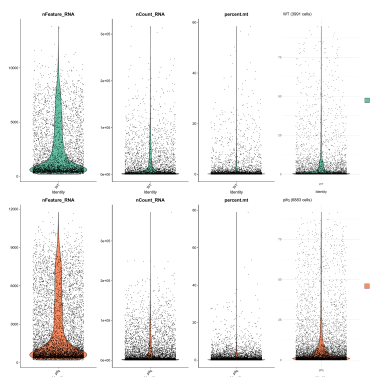


Figure 1: Pre-filter QC violin plots.

2.4 Section 2 - Cell filtering and doublet detection

This section filters low-quality cells and then runs doublet detection. The code shown here uses two thresholds: cells must have more than 200 detected genes (`min_features = 200`) and less than 5% mitochondrial reads (`max_mt = 5`).

The filtering function also accepts extra variables that are not used explicitly in this example:

- `max_features`: maximum number of detected genes per cell.
- `min_counts`: minimum total RNA counts per cell.
- `max_counts`: maximum total RNA counts per cell.
- `max_cp`: maximum chloroplast percentage.
- `run_doubletfinder`: whether DoubletFinder should be run after filtering.

These extra options are useful when the QC plots show cells with very low complexity, unusually high gene or UMI counts, high chloroplast signal, or when you want to disable doublet detection for a quick test.

After running the block, the pipeline saves a post-filter QC plot in `results/01_qc/qc_postfilter.pdf` and stores the filtered sample list in `results/objects/seurat_list_postfilter.rds`. This saved object is used by the next section.

R

```
output_dir <- dir_01
seurat_list <- filter_seurat_samples(seurat_list_raw, min_features = 200, max_mt = 5)
plot_qc_batch(seurat_list, colors, "qc_postfilter.pdf")
saveRDS(seurat_list, file.path(dir_objects, "seurat_list_postfilter.rds"))
message("\nOK SECTION 2 COMPLETE: filtering and doublet detection complete")
```

Representative output from Section 2:

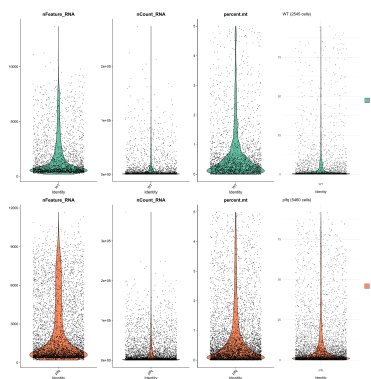


Figure 2: Post-filter QC violin plots.

2.5 Section 3 - Merge and initial preprocessing

This section combines the filtered samples into one Seurat object and performs the first standard preprocessing steps. At this point, the samples are merged but not yet batch-corrected.

The workflow normalizes the data, identifies informative genes, scales the expression matrix, summarizes the main sources of variation with PCA, and creates a first UMAP visualization. This UMAP is useful for checking how the samples look before Harmony correction.

The output is saved as `results/objects/ath_sc_preharmony.rds`, and the UMAP before batch correction is saved as `results/01_qc/umap_preharmony.pdf`.

R

```
output_dir <- dir_01
ath_sc <- reduce(seurat_list, merge) %>%
  NormalizeData(verbose = FALSE) %>%
  FindVariableFeatures(
    selection.method = "vst", nfeatures = 2000, verbose = FALSE
  ) %>%
  ScaleData(verbose = FALSE) %>%
  RunPCA(npcs = 30, verbose = FALSE) %>%
  RunUMAP(reduction = "pca", dims = 1:30, verbose = FALSE)
save_pdf(
  DimPlot(ath_sc, group.by = "orig.ident", cols = colors),
  "umap_preharmony.pdf"
)
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_preharmony.rds"))
message("\nOK SECTION 3 COMPLETE: merge and preprocessing complete")
```

Representative output from Section 3:

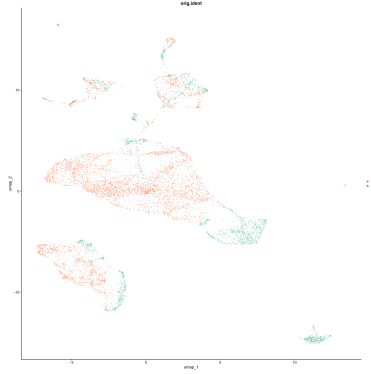


Figure 3: UMAP before Harmony correction, colored by sample.

2.6 Section 4 - Harmony batch correction

This section applies Harmony to reduce sample-specific batch effects after the initial merge. The correction is done using the sample identity stored in `orig.ident`.

After Harmony, the UMAP is recalculated using the corrected representation. This second UMAP helps check whether samples mix better while still preserving biological structure.

The corrected object is saved as `results/objects/ath_sc_postharmony.rds`, and the post-Harmony UMAP is saved as `results/01_qc/umap_postharmony.pdf`.

R

```
output_dir <- dir_01
ath_sc <- ath_sc %>%
  RunHarmony("orig.ident", plot_convergence = FALSE) %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE)
save_pdf(
  DimPlot(ath_sc, group.by = "orig.ident", cols = colors),
  "umap_postharmony.pdf"
)
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_postharmony.rds"))
message("\nOK SECTION 4 COMPLETE: Harmony integration complete")
```

Representative output from Section 4:

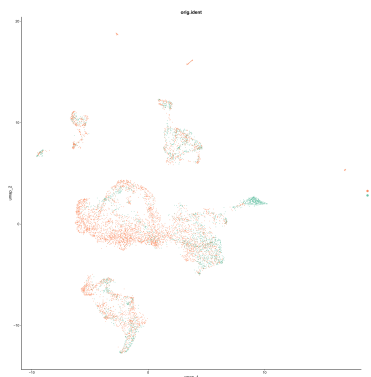


Figure 4: UMAP after Harmony correction, colored by sample.

2.7 Section 5 - Resolution optimization

This section helps choose a reasonable clustering resolution before assigning the final clusters. The `resolutions_test` vector contains the candidate resolutions that will be tested. Lower values produce fewer, broader clusters; higher values produce more, smaller clusters.

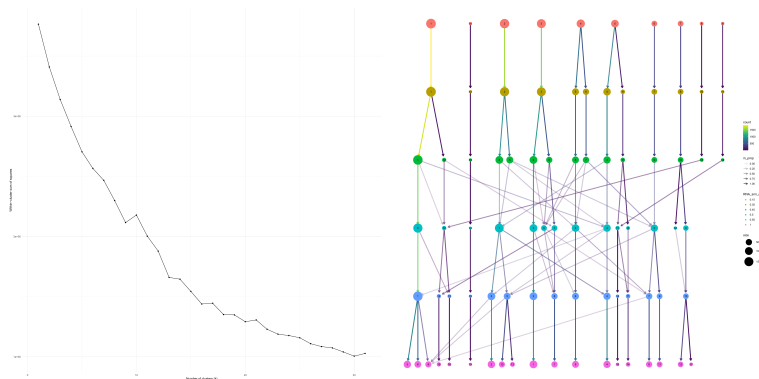
The section generates two diagnostic plots: an elbow plot and a clustree. The elbow plot gives a broad view of how many groups are supported by the data. The clustree shows how clusters split or remain stable across the tested resolution values. Together, these plots help avoid choosing a resolution that is too coarse or too fragmented.

The outputs are saved in `results/02_clustering/` as `elbow_plot.pdf` and `clustree2.pdf`. The underlying PCA dimensions, k-means settings, and neighbor count are fixed inside `plot_resolution_elbow()` and `run_resolution_sweep()` (`workflow/ScRNA_Analysis_Functions.R`) and rarely need changing; the only thing to edit here is `resolutions_test`.

R

```
resolutions_test <- c(0.15, 0.35, 0.45, 0.55, 1.0)
output_dir <- dir_02
plot_resolution_elbow(ath_sc, output_dir = output_dir)
clu <- run_resolution_sweep(ath_sc, resolutions = resolutions_test, output_dir = output_dir)
message("\nOK SECTION 5 COMPLETE: elbow plot and clustree saved")
```

Representative outputs from Section 5:



2.8 Section 6 - Final clustering

This section applies the final clustering resolution chosen after inspecting the diagnostic plots from Section 5. The value is set in `cluster_resolution`.

To use this block, first look at the elbow plot and clustree from Section 5, then choose the resolution that gives a reasonable number of stable clusters. Lower values produce fewer, broader clusters; higher values produce more, smaller clusters.

After running this block, each cell receives a final Seurat cluster ID stored in `seurat_clusters`. These numeric clusters are the starting point for cell-type annotation in the next section.

The output figure is a UMAP colored by final cluster number, saved as `results/02_clustering/umap_seuratclusters.pdf`.

R

```
cluster_resolution <- 0.35
output_dir <- dir_02
ath_sc <- ath_sc %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(reduction = "harmony", dims = 1:30, k.param = 20, verbose = FALSE) %>%
  FindClusters(resolution = cluster_resolution, algorithm = 4, verbose = FALSE)
Idents(ath_sc) <- "seurat_clusters"
save_pdf(
  DimPlot(ath_sc, group.by = "seurat_clusters", label = TRUE),
  "umap_seuratclusters.pdf"
)
message("\nOK SECTION 6 COMPLETE: final clustering complete")
```

Representative output from Section 6:

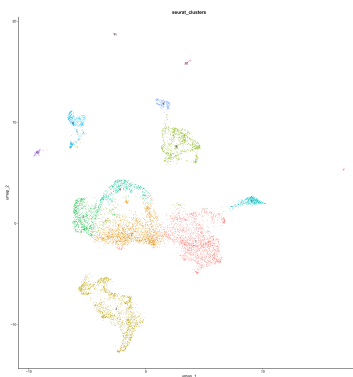


Figure 5: Final UMAP colored by numbered Seurat clusters.

2.9 Section 7 - Cell-type annotation

This section assigns biological cell-type labels to the clusters using a marker gene table from the literature. The marker file is read from `data/biblio_marks.txt`.

The pipeline first finds marker genes for the dataset, then compares those markers with the bibliography table to assign a `celltype` label. The dotplot is used to inspect whether the expected marker genes support the assigned labels.

The marker table is saved in `results/03_annotation/FindAllMarkers.tsv`, and the main figures are saved as `dotplot_marker_table_annotation_biblio.pdf` and `umap_annotation_biblio.pdf`.

R

```
output_dir <- dir_03
biblio_marks_file <- file.path(DATA_DIR, "data", "biblio_marks.txt")
marker_table <- read.table(
  biblio_marks_file, header = TRUE, sep = "\t", quote = ""
)
markers <- find_markers(
  ath_sc,
  output_file = file.path(output_dir, "FindAllMarkers.tsv")
)
ath_sc <- annotate_by_markers(
  ath_sc, markers,
  reference_file = biblio_marks_file
)
plot_marker_dotplot(
  ath_sc,
  marker_table,
  annot_col = "celltype",
  outfile = file.path(
    output_dir, "dotplot_marker_table_annotation_biblio.pdf"
  ),
  width = 18, height = 18
)
save_pdf(
  DimPlot(
    ath_sc, group.by = "celltype",
    label = TRUE, repel = TRUE, raster = FALSE
  ),
  "umap_annotation_biblio.pdf"
)
message("\nOK SECTION 7 COMPLETE: cell-type annotation complete")
```

Representative outputs from Section 7:

2.10 Section 8 - Annotated clustree

This section adds the cell-type labels from Section 7 onto the clustering tree from Section 5. Instead of showing only numeric clusters, each node is labeled with the most common cell type found in that cluster.

This makes it easier to see how stable the annotated cell types are across different clustering resolutions. The output is saved as `results/03_annotation/clustree_annotated.pdf`.

R

```
output_dir <- dir_03
plot_annotated_clustree(ath_sc, clu, annot_col = "celltype", output_dir = output_dir)
message("\nOK SECTION 8 COMPLETE: annotated clustree saved")
```

Representative output from Section 8:

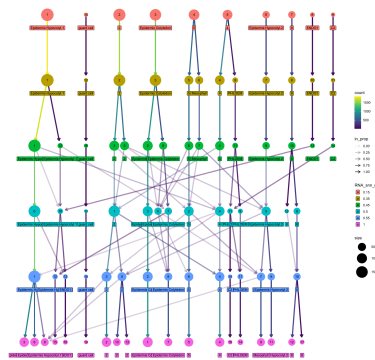


Figure 6: Cluster tree labeled by the dominant bibliography-based annotation in each node.

2.11 Section 9 - Gene expression visualization

This section plots the expression of one selected gene across the annotated cell types. The gene to inspect is set in the `gene` variable.

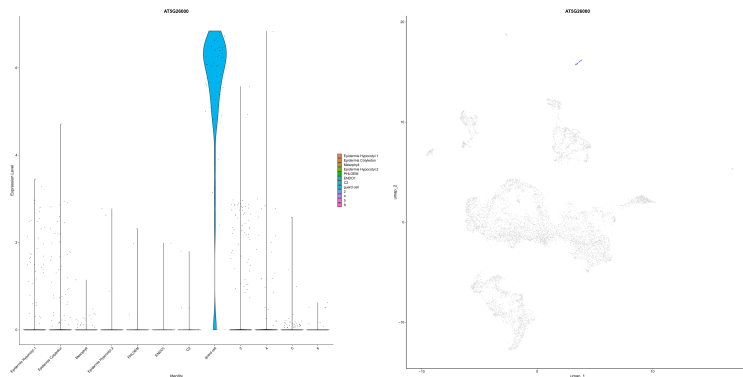
The violin plot shows expression levels by cell type, while the FeaturePlot shows where the same gene is expressed on the UMAP. These two views are useful for checking whether a marker or gene of interest matches the expected cell population.

The outputs are saved in `results/04_expression/` with filenames based on the selected gene.

R

```
gene <- "AT5G26000"
output_dir <- dir_04
ath_sc <- JoinLayers(ath_sc)
Idents(ath_sc) <- "celltype"
gene_tag <- paste(gene, collapse = "_")
save_vln(
  VlnPlot(ath_sc, features = gene),
  paste0("vln_gene_", gene_tag, ".pdf")
)
save_pdf(
  FeaturePlot(ath_sc, features = gene),
  paste0("feature_gene_", gene_tag, ".pdf")
)
message("\nOK SECTION 9 COMPLETE: gene expression visualization saved")
```

Representative outputs from Section 9:



2.12 Section 10 - Cell-type grouping [optional]

This optional section simplifies annotation labels by merging selected cell types or subclusters into broader groups. It uses the `grouping` object as a small renaming table: the names on the left are the current labels, and the values on the right are the new labels.

For example, in this report `Epidermis Hypocotyl.1`, `.2`, and `.3` are all renamed as `Epidermis Hypocotyl`. Any cell type that is not listed in `grouping` keeps its original label.

Use this section when the detailed annotation is useful for inspection but a simpler grouping is better for figures or downstream summaries. The grouped labels are stored in `celltype_grouped`, and the grouped UMAP is saved as `results/05_curation/umap_grouped.pdf`.

R

```
# Merge the three epidermis hypocotyl subclusters into a single label
grouping <- c(
  "Epidermis Hypocotyl.1" = "Epidermis Hypocotyl",
  "Epidermis Hypocotyl.2" = "Epidermis Hypocotyl",
  "Epidermis Hypocotyl.3" = "Epidermis Hypocotyl"
)

output_dir <- dir_05

if (length(grouping) > 0) {
  ath_sc$celltype_grouped <- recode(ath_sc$celltype, !!!grouping)
} else {
  ath_sc$celltype_grouped <- ath_sc$celltype
}

save_pdf(
  DimPlot(
    ath_sc, group.by = "celltype_grouped",
    label = TRUE, repel = TRUE, raster = FALSE
  ),
  "umap_grouped.pdf"
)

message("\nOK SECTION 10 COMPLETE: cell-type grouping complete")
```

Representative output from Section 10:

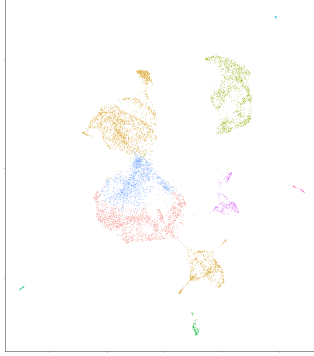


Figure 7: UMAP colored by grouped cell-type labels.

2.13 Section 11 - Subcluster marker inspection [optional]

This optional section is for manual curation when one annotated cluster needs a closer look. First, `inspect_subcluster_markers()` takes one cluster, splits it into smaller subclusters, and saves a figure with the subcluster UMAP and marker gene patterns. This helps decide whether the original label should be kept or refined.

If refinement is needed, `reassign` is edited as a small mapping table. The outer name is the cluster being curated, the names inside are subcluster IDs, and the values are the new labels. The special label `others` is used for any subcluster not listed explicitly.

Then `curate_clusters()` applies those manual labels and stores the result in `celltype_curated`. The final curated UMAP is saved as `results/05_curation/umap_curated.pdf`. This block is not executed automatically because the label choices depend on inspecting the figures.

R

```
output_dir <- dir_05
Idents(ath_sc) <- "celltype"

# 1. Inspect a cluster (saves the combined subcluster + marker figure)
inspect_subcluster_markers(
  ath_sc, cluster_id = "Mesophyll",
  marker_table = marker_table, output_dir = output_dir
)

# 2. Map each cluster's subclusters to labels ("others" = any ID not listed),
# then curate them in one call. Here Mesophyll's subclusters all map back
# to "Mesophyll" (no relabeling), which is the pattern to use when the
# subcluster inspection confirms the original label was already correct:
reassign <- list(
  "Mesophyll" = c("others" = "Mesophyll")
)
ath_sc <- curate_clusters(
  ath_sc, reassign,
  marker_table = marker_table, output_dir = output_dir
)

# 3. Save the curated annotation UMAP
save_pdf(
  DimPlot(ath_sc, group.by = "celltype_curated",
    label = TRUE, repel = TRUE, raster = FALSE),
  "umap_curated.pdf"
)

message("\nOK SECTION 11 COMPLETE: curation complete")
```

Representative output from Section 11 — subcluster inspection of the Mesophyll population (UMAP of its subclusters plus marker-gene panels used to decide whether to keep or split the label):

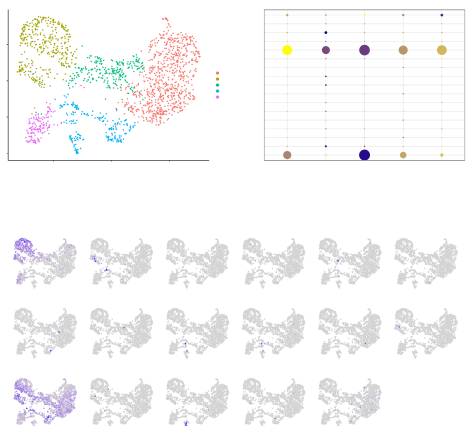


Figure 8: Mesophyll subcluster inspection: subcluster UMAP, marker dotplot, and per-gene marker panels.

This example was executed in a validation run of this pipeline on the bibliography-annotated Mesophyll population. Its subclusters were mapped back to `others = "Mesophyll"` (no relabeling), which is why `celltype_curated` matches `celltype` for Mesophyll in that run — this demonstrates the inspect-then-curate mechanism without changing any label. The rest of this document (from Section 13 on) uses the reference run's `celltype_curated` labels shown in the table below, which came from curating its own unresolved clusters (1, 2, 8, 10, 11) individually.

2.14 Section 12 - Export the curated object

This section closes the R part of the workflow by saving the final object in two formats. The `.rds` file is used to continue working in R, and the `.h5ad` file is used by the Python sections of the pipeline.

The outputs are saved in `results/objects/` as `ath_sc_curated.rds` and `ath_sc_curated.h5ad`.

R

```
# Checkpoint - restore with:
#   ath_sc <- readRDS(file.path(dir_objects, "ath_sc_curated.rds"))
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_curated.rds"))

# Export the curated object to AnnData h5ad format for Python-based
# trajectory and velocity analyses.
export_to_scanpy(
  ath_sc,
  file.path(dir_objects, "ath_sc_curated.h5ad")
)

# To export a specific cell type only:
# export_to_scanpy(
#   subset(ath_sc, subset = celltype_curated == "GC"),
#   file.path(dir_objects, "GuardCell.h5ad")
# )

message("\nOK SECTION 12 COMPLETE: curated object saved (.rds + .h5ad)")
```

3 Part 3 - Pseudobulk differential expression and GO

This part continues from the object saved at the end of Chapter 1. It aggregates cells into pseudo-replicates per cell type, runs pseudobulk DESeq2 for the WT vs pifq contrast, and summarizes results with volcano plots, differential tables, GO enrichment, a log2FC heatmap, and co-expression networks.

R

```
ath_sc <- readRDS(file.path(dir_objects, "ath_sc_curated.rds"))
```

3.1 Section 13 - Cell-type subsets

The annotation column used to define cell types is a user choice. Depending on how Chapter 1 was run, this is one of `celltype` (bibliography annotation), `celltype_grouped` (broad groups from Section 10), or `celltype_curated` (manual curation from Section 11). All downstream sections in this part use whichever column is set here.

R

```
pseudobulk_annot_col <- "celltype_curated"
table(ath_sc[[pseudobulk_annot_col]])
cell_type_subsets <- create_cell_type_subsets(
  ath_sc, annot_col = pseudobulk_annot_col
)
message("\nOK SECTION 13 COMPLETE: cell-type subsets created")
```

Cells per cell type (bibliography annotation `celltype_curated`; the two mesophyll subclusters and the still-numeric unannotated clusters are kept separate at this stage):

Cell type	Cells
Epidermis	1464
SAM	875
Mesophyll_1	680
Mesophyll_2	489
MMC	366
Vascular	148
GC	49
1	1839
2	1594
8	366
10	78
11	57

3.2 Section 14 - Pseudo-replicate assignment

This section creates pseudo-replicates for the pseudobulk analysis. Instead of treating each single cell as an independent replicate, cells from the same cell type and condition are split into a small number of groups. Each group becomes one pseudo-replicate.

The number of pseudo-replicates is controlled by `n_pseudoreps`. In this report, `n_pseudoreps = 3`, so each condition inside each cell type is split into three pseudo-replicates when enough cells are available.

`pseudobulk_conditions` can be left as `NULL` to use all conditions present in the object, or it can be set to specific conditions if only some groups should be used. Cell types that do not contain enough information for a comparison, such as cell types present in only one condition, are removed from the pseudobulk contrast.

The output is `cell_type_subsets_replicates`, which is the same set of cell-type subsets but with pseudo-replicate labels added. This object is used in the next section to build pseudobulk count tables.

R

```
pseudobulk_conditions <- NULL
n_pseudoreps          <- 3
cell_type_subsets_replicates <- assign_pseudoreplicates_batch(
  cell_type_subsets,
  pseudobulk_conditions = pseudobulk_conditions,
  n_pseudoreps          = n_pseudoreps
)
table(cell_type_subsets_replicates[["Epidermis"]]$replicate)
message("\nOK SECTION 14 COMPLETE: pseudo-replicates assigned")
```

Cell types present in only one condition (here cluster 11, with only 57 cells and all from the pifq sample) are dropped, since a WT vs pifq contrast needs both conditions represented. Eleven cell types are retained.

Cells per pseudo-replicate (first two cell types shown):

Cell type	Replicate	Cells
Epidermis	pifq_rep1	318
Epidermis	pifq_rep2	353
Epidermis	pifq_rep3	329
Epidermis	WT_rep1	164
Epidermis	WT_rep2	155
Epidermis	WT_rep3	145
Mesophyll_1	pifq_rep1	223
Mesophyll_1	pifq_rep2	218
...	...	...

3.3 Section 15 - Pseudobulk tables and DESeq2

This section builds pseudobulk count tables and runs DESeq2 for the selected comparisons. For each cell type, counts from cells belonging to the same pseudo-replicate are summed together, creating replicate-level count profiles that can be analyzed with a bulk RNA-seq method.

The comparisons are defined in the `comparisons` list. Each entry contains the two conditions to compare and a short `tag` used to name the output folder and files. In this report, the comparison is WT versus pifq, and the tag is `WT_vs_pifq`.

`cell_types_to_analyze` can be left as `NULL` to analyze all available cell types, or set to a selected list if only specific cell types should be tested.

For each cell type and comparison, the pipeline writes a DESeq2 result table with one row per gene. The main columns are `log2FoldChange`, `pvalue`, and `padj`. For the `WT_vs_pifq` comparison used here, positive `log2FC` means higher expression in pifq, while negative `log2FC` means higher expression in WT.

R

```
comparisons <- list(
  list(conds = c("WT", "pifq"), tag = "WT_vs_pifq")
)
cell_types_to_analyze <- NULL
output_dir <- dir_06
deseq2_results <- run_pseudobulk_deseq2_analysis(
  cell_type_subsets_replicates = cell_type_subsets_replicates,
  comparisons = comparisons,
  output_dir = output_dir,
  cell_types = cell_types_to_analyze,
  pseudobulk_dir = file.path(dir_objects, "pseudobulk_replicates")
)
message("\nOK SECTION 15 COMPLETE: pseudobulk aggregation and DESeq2 complete")
```

One `DESeq2_<cell_type>_WT_vs_pifq.csv` is written per cell type: the standard DESeq2 results table (one row per gene, with `log2FoldChange`, `pvalue`, and `padj`). Positive `log2FC` means higher expression in `pifq`. First rows of the `Mesophyll_1` table:

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
AT1G01010	16.74	-1.651	0.418	-3.946	7.9e-05	9.3e-04
AT1G01020	27.42	0.697	0.388	1.798	7.2e-02	2.2e-01
AT1G01030	117.17	-1.439	0.501	-2.871	4.1e-03	2.6e-02

3.4 Section 16 - Volcano plots

This section creates volcano plots from the DESeq2 result tables generated in Section 15. Each plot summarizes one cell type for the selected comparison.

The `volcano_tag` selects which comparison folder to use. The thresholds `padj_cut` and `lfc_cut` define which genes are highlighted as significant in the plots. In this report, genes are highlighted when adjusted p-value is below 0.05 and absolute `log2FC` is at least 1.

The volcano plots are saved inside the corresponding comparison folder under `results/06_de_results/`.

R

```
volcano_tag <- "WT_vs_pifq"
padj_cut <- 0.05
lfc_cut <- 1
render_volcano_plots(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir = file.path(dir_06, volcano_tag, "volcano"),
  pdf_name = paste0("VolcanoPlots_", volcano_tag, ".pdf"),
  padj_cut = padj_cut,
  lfc_cut = lfc_cut
)
message("\nOK SECTION 16 COMPLETE: volcano plots saved")
```

Representative output from Section 16:



Figure 9: Volcano plot for Mesophyll_1, WT vs pifq.

3.5 Section 17 - Differential gene tables

This section summarizes the DESeq2 results into combined differential gene tables. It uses the same significance thresholds defined for the volcano plots: `padj_cut` for adjusted p-value and `lfc_cut` for log2 fold-change.

The output makes it easier to compare differential expression across cell types. One table records whether each gene is significantly upregulated, downregulated, or not significant in each cell type. A second table stores the corresponding log2FC values for the significant genes.

These tables are useful for quickly checking which genes are shared across cell types and which responses are cell-type specific.

R

```
diff_prefix <- paste0("diff_table_", volcano_tag)
diff_tables <- build_differential_tables(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut,
  prefix      = diff_prefix
)
message("\nOK SECTION 17 COMPLETE: differential gene tables saved")
```

This writes two combined matrices (genes x cell types). The class matrix (`diff_table_WT_vs_pifq_*.tsv`) uses +1/-1/0 to flag genes up-, down-, or not significantly regulated in each cell type; the log2FC matrix (`log2FC_table_*.tsv`) holds the actual fold changes (NA where not significant). Example rows for three annotated cell types (Epidermis, Mesophyll_1, SAM):

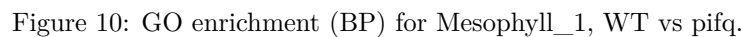
# class matrix				# log2FC matrix			
gene_id	Epid	Meso1	SAM	gene_id	Epid	Meso1	SAM
AT1G01010	0	-1	-1	AT1G01010	NA	-1.65	-1.51
AT1G01030	0	-1	1	AT1G01030	NA	-1.44	3.05
AT1G01060	1	1	0	AT1G01060	2.35	4.08	NA

3.6 Section 18 - GO enrichment

This section tests whether the significant genes from each cell type are enriched for Gene Ontology terms. In simple terms, it asks whether the genes that change between conditions are concentrated in particular biological processes, functions, or cellular components.

The adjusted p-value cutoff is controlled by `padj_cutoff`. The GO result tables and bubble plots are saved in `results/07_go/` under the selected comparison name.

Representative output from Section 18:



This section turns the differential expression tables into a heatmap of log2FC values. It is used to compare how strongly genes change across cell types in the same condition contrast.

The heatmap is saved in the comparison folder inside `results/06_de_results/`.

R

```
heatmap_limits <- c(-5, 5)
build_logfc_heatmap(
  logfc_table = diff_tables$logfc,
  contrast_tag = volcano_tag,
  output_dir = file.path(dir_06, volcano_tag),
  limits = heatmap_limits,
  cell_order = unique(marker_table$cell.types)
)
message("\nOK SECTION 19 COMPLETE: log2FC heatmap saved")
```

Representative output from Section 19:

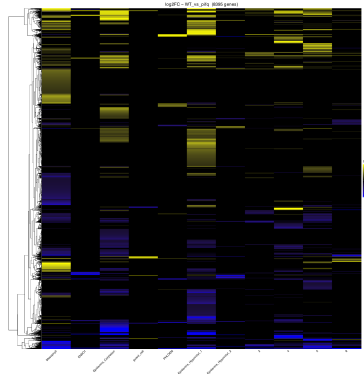


Figure 11: log2FC heatmap: rows hierarchically clustered, columns in marker-table order (unannotated clusters last), WT vs pifq.

3.8 Section 20 - Gene-gene coexpression and TF network

This section builds a gene-gene co-expression network from the significant differentially expressed genes, then focuses the network on transcription factor connections. The transcription factor list comes from `data/AtTFDB_loci.txt`, which was prepared from AtTFDB, the Arabidopsis transcription factor database available through AGRIS: <https://agris-knowledgebase.org/>.

The main input is the log2FC table produced in Section 17. The function uses the Seurat object, the differential expression results, and the annotation column to build metacells, estimate co-expression, keep strong gene-gene edges, and draw a network colored by differential expression direction.

The parameters near the end of the function call control practical details such as the number of metacells, the minimum module size, the edge-strength threshold, and how many hub transcription factors are labeled. The network output is saved in `results/08_networks/`.

R

```
run_tf_coexpression_network(  
  seurat_obj      = ath_sc,  
  de_table_path   = file.path(dir_06, volcano_tag, "log2FC_table_fc1_padj_005.tsv"),  
  tf_list_path    = file.path(DATA_DIR, "data", "AtTFDB_loci.txt"),  
  output_dir      = dir_08,  
  annot_col       = pseudobulk_annot_col,  
  contrast_tag    = volcano_tag,  
  n_metacells     = 50,  
  min_module_size = 20,  
  deep_split      = 2,  
  soft_power      = NULL,  
  tom_threshold   = 0.2,  
  n_hub_label     = 15  
)  
message("\nOK SECTION 20 COMPLETE: TF coexpression network saved")
```

Representative output from Section 20:

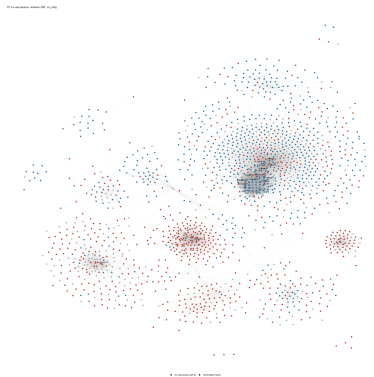


Figure 12: TF co-expression network, colored by DE direction.

4 Part 4 - Pseudotime trajectory analysis (Python)

Chapter 3 runs in Python (scanpy, Palantir, scFates), starting from the h5ad object exported at the end of Chapter 1 (results/objects/ath_sc_curated.h5ad). Exit R and start Python inside the same Docker container:

Bash

```
q()          # leave R (do not save the workspace)
python3
```

4.1 Section 23 - Setup

This section starts the Python part of the pipeline. It defines the same project paths used earlier, loads the Python helper scripts, and creates the pseudotime output folder.

Use this block first after leaving R and starting Python inside the Docker container. After it runs, the functions needed for trajectory analysis are available in the Python session.

Python

```
import os
PIPELINE_DIR = "/workspace/workflow"
DATA_DIR     = "/workspace/."
base_dir     = os.path.join(DATA_DIR, "results")
exec(open(os.path.join(PIPELINE_DIR, "load_libraries_python.py")).read(), globals())
exec(open(os.path.join(PIPELINE_DIR, "ScRNA_Pseudotime_Functions.py")).read(), globals())
dir_pseudotime = os.path.join(base_dir, "09_pseudotime")
os.makedirs(dir_pseudotime, exist_ok=True)
print("SECTION 23 COMPLETE: setup done")
```

4.2 Section 24 - Load curated object

This section loads the .h5ad object exported at the end of the R workflow. The INPUT_H5AD path points to that file, and ANNOTATION_COL selects which cell type annotation column will be used for plotting and selecting cells.

The function also standardizes coordinate names so the object works cleanly with Scanpy. It saves an overview UMAP and prints the available cell types, which are used to choose the trajectory cells in the next section.

Python

```
INPUT_H5AD      = "/workspace/results/objects/ath_sc_curated.h5ad"
ANNOTATION_COL  = "celltype_curated"
N_JOBS          = 4
adata, N_JOBS = load_curated_object(
    input_h5ad      = INPUT_H5AD,
    dir_pseudotime  = dir_pseudotime,
    annotation_col  = ANNOTATION_COL,
    n_jobs          = N_JOBS,
)
```

Representative output from Section 24:

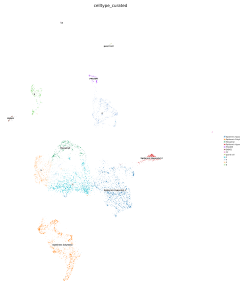


Figure 13: Overview UMAP colored by curated cell type.

4.3 Section 25 - Cell type selection

This section selects which cell types will be used for trajectory inference. The list is set in `TRAJECTORY_CLUSTERS`, and the names must match the cell-type labels printed in Section 24.

In this report, only `epidermis cotyledon` is selected. The preview UMAP shows these cells before fitting the trajectory, which helps confirm that the intended population was chosen.

Python

```
TRAJECTORY_CLUSTERS = ["epidermis cotyledon"]
adata_sub = preview_trajectory_selection(
    adata = adata,
    clusters = TRAJECTORY_CLUSTERS,
    annotation_col = ANNOTATION_COL,
    dir_pseudotime = dir_pseudotime,
)
```

Representative output from Section 25:

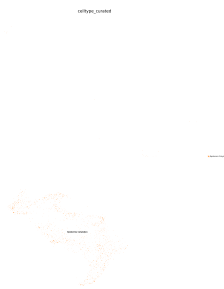


Figure 14: UMAP of the epidermis cotyledon cells selected for the trajectory.

4.4 Section 26 - Trajectory inference

This section fits the pseudotime trajectory for the selected cells. The `ROOT_CLUSTER` defines where pseudotime starts. Here, the root is set to `epidermis cotyledon`, so the trajectory is oriented within that selected

population.

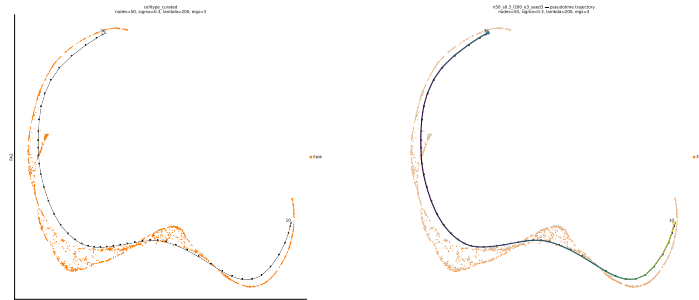
TRAJECTORY_RUNS stores the trajectory settings to test. Each run creates its own output folder, making it possible to compare parameter choices without overwriting previous results. The run shown here uses 50 nodes, $\sigma = 0.3$, $\lambda = 200$, $eigs = 3$, and $seed = 3$.

Here, $eigs = 3$ means that three diffusion components are used before fitting the trajectory. The outputs include the trajectory graph, pseudotime visualization, and selected root cell plot inside `results/09_pseudotime/`.

Python

```
ROOT_CLUSTER = "epidermis cotyledon"
TRAJECTORY_RUNS = [
    trajectory_run(nodes=50, sigma=0.3, lambda_value=200, eigs=3, seed=3),
]
adata_traj, selected_trajectory_dir, trajectory_runs = run_trajectory_runs(
    adata          = adata,
    clusters       = TRAJECTORY_CLUSTERS,
    root_cluster   = ROOT_CLUSTER,
    annotation_col = ANNOTATION_COL,
    output_base_dir = dir_pseudotime,
    runs           = TRAJECTORY_RUNS,
)
```

Representative outputs from Section 26 for the epidermis-cotyledon-only trajectory using `nodes=50`, $\sigma=0.3$, $\lambda=200$, and $eigs=3$:



4.5 Section 27 - Plot genes on trajectory

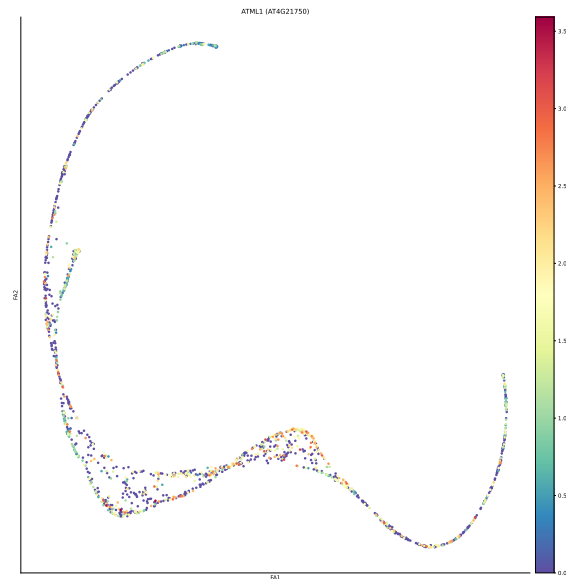
This section plots selected genes on the fitted epidermis cotyledon trajectory. The gene list is controlled by GENES, so changing that vector is the only thing needed to inspect another marker.

For this version, Section 27 is focused only on ATML1. In the count matrix ATML1 is stored as the locus AT4G21750, so the code uses AT4G21750 while the text and figure label report it as ATML1. The plot shows where ATML1 expression falls on the epidermis cotyledon trajectory layout.

Python

```
GENES = ["AT4G21750"]
name = os.path.basename(selected_trajectory_dir)
ensure_expression_in_X(adata_traj)
gene_plots_dir = os.path.join(selected_trajectory_dir, "gene_plots")
os.makedirs(gene_plots_dir, exist_ok=True)
fig = sc.pl.draw_graph(
    adata_traj,
    color=GENES,
    title=["ATML1 (AT4G21750)"],
    show=False,
    return_fig=True,
)
save_plot_18x18(fig, os.path.join(gene_plots_dir, f"{name}_gene_plots.pdf"))
plt.close(fig)
print("SECTION 27 COMPLETE: gene trajectory plots saved")
```

Representative output from Section 27:



ATML1 (AT4G21750) expression on the epidermis cotyledon trajectory.

4.6 Section 28 - Gene trends

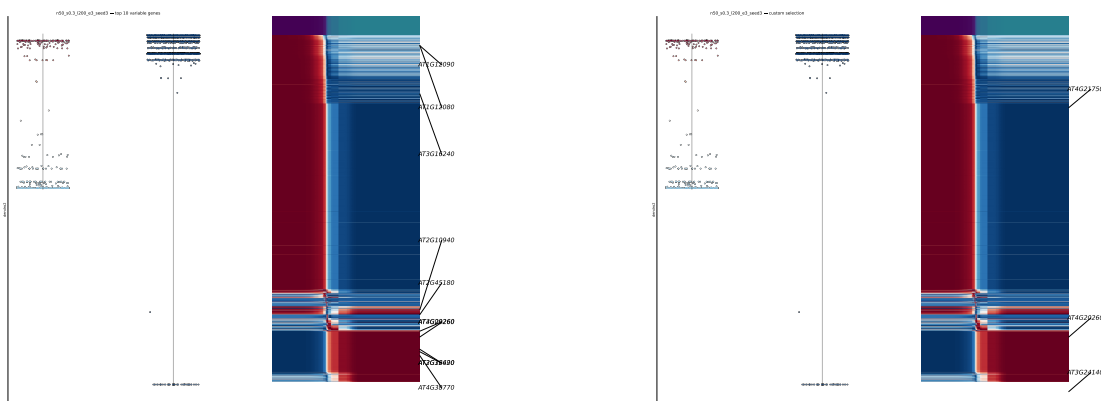
This section fits gene-expression trends along pseudotime for the epidermis cotyledon trajectory calculated above. The function first tests genes for association with pseudotime, fits smooth expression curves, ranks genes by their trend signal, and then saves trend heatmaps.

For this report, the run shown here uses the already selected epidermis cotyledon trajectory: `nodes=50`, `sigma=0.3`, `lambda=200`, `eigs=3`, and `seed=3`. `TOP_N = 10` means the main panel shows the top 10 ranked genes. `HIGHLIGHT_GENES = ["AT3G24140"]` highlights the selected marker using the same fitted trend workflow.

Python

```
TOP_N          = 10
HIGHLIGHT_GENES = ["AT3G24140"]
ORDERING       = "max"
SPLINE_DF      = 3
name = os.path.basename(selected_trajectory_dir)
ensure_expression_in_X(adata_traj)
top_path, highlight_path = run_step29_gene_trends(
    adata      = adata_traj,
    run_dir    = selected_trajectory_dir,
    name       = name,
    custom_genes = HIGHLIGHT_GENES,
    top_n      = TOP_N,
    ordering   = ORDERING,
    n_jobs     = N_JOBS,
    spline_df  = SPLINE_DF,
)
print("SECTION 28 COMPLETE: epidermis cotyledon gene trends saved")
```

Representative outputs from Section 28 — epidermis cotyledon gene trends:



Left: top 10 ranked genes along pseudotime for the epidermis cotyledon trajectory. Right: AT3G24140 highlighted from the same fitted trend workflow.

4.7 Section 29 - Final export

This section saves the final pseudotime object after trajectory inference and gene-trend analysis. The exported `.h5ad` file keeps the trajectory results, pseudotime values, and layout coordinates for later reuse.

The final object is saved in `results/objects/` as `ath_sc_pseudotime_final.h5ad`.

Python

```
final_h5ad = os.path.join(base_dir, "objects", "ath_sc_pseudotime_final.h5ad")
os.makedirs(os.path.dirname(final_h5ad), exist_ok=True)
adata_traj.write_h5ad(final_h5ad)
print(f"SECTION 29 COMPLETE: final pseudotime object saved -> {final_h5ad}")
```

Supplementary Table — Software dependencies and versions

Package versions are those bundled in the `psblab/scrnaseq:latest` Docker image (the public image used for all R and Python steps). R **base-priority** packages are omitted. Cell Ranger is 10x Genomics proprietary software run *outside* the container (Part 1); its version is fixed by the workflow.

Interpreters and command-line tools

Tool	Version
Cell Ranger	9.0.1
R	4.5.3
Python	3.12.3

Installing tmux or screen

OS	tmux	screen
Linux (Debian/Ubuntu)	<code>sudo apt install tmux</code>	<code>sudo apt install screen</code>
Linux (RHEL/CentOS/Fedora)	<code>sudo yum install tmux</code>	<code>sudo yum install screen</code>
macOS (Homebrew)	<code>brew install tmux</code>	ships with macOS by default
Windows	not natively available — see note below	not natively available — see note below

Windows users: this whole pipeline (Cell Ranger, Docker/Singularity, R, Python) is Linux-based. The practical options on Windows are WSL2 (<https://learn.microsoft.com/windows/wsl/install>) running Ubuntu locally, or an SSH client (Windows Terminal, PuTTY, or WSL2's own `ssh`) connecting out to a Linux server or HPC cluster where the rest of this guide applies directly. Once inside WSL2 or an SSH session on Linux, use the Linux commands above.

Installing a container runtime (Docker or Singularity/Apptainer)

Pick one; both run the same `psblab/scrnaseq:latest` image (see “Container startup” in Part 1).

Docker

OS	Install
Linux	https://docs.docker.com/engine/install/ (distro-specific packages)
macOS	Docker Desktop: https://docs.docker.com/desktop/install/mac-install/
Windows	Docker Desktop with the WSL2 backend: https://docs.docker.com/desktop/install/windows-install/

Singularity / Apptainer (no root required; common on HPC clusters where Docker is not permitted)

OS	Install
Linux	https://apptainer.org/docs/admin/main/installation.html
macOS / Windows	not natively supported — use a Linux VM, WSL2, or the remote server/cluster itself

On a shared HPC cluster, Singularity/Apptainer is usually already installed system-wide — check with `singularity --version` or `apptainer --version` before installing anything.

R packages (421)

Package	Version	Package	Version
abind	1.4-8	locfit	1.5-9.12
anndataR	1.0.2	lubridate	1.9.5
AnnotationDbi	1.72.0	magrittr	2.0.5
ape	5.8-1	maps	3.4.3
aplot	0.2.9	MASS	7.3-65
askpass	1.2.1	Matrix	1.7-5
assertthat	0.2.1	MatrixGenerics	1.22.0
assorthead	1.4.0	MatrixModels	0.5-4
backports	1.5.1	matrixStats	1.5.0
base64enc	0.1-6	memoise	2.0.1
basilisk	1.22.0	mgcv	1.9-4
batchelor	1.26.0	microbenchmark	1.5.0
beachmat	2.26.0	mime	0.13
beeswarm	0.4.0	miniUI	0.1.2
BH	1.90.0-1	minqa	1.2.8
biglm	0.9-3	modelr	0.1.11
Biobase	2.70.0	monocle3	1.4.27
BiocGenerics	0.56.0	nlme	3.1-169
BiocManager	1.30.27	nloptr	2.2.1
BiocNeighbors	2.4.0	nnet	7.3-20
BiocParallel	1.44.0	numDeriv	2016.8-1.1
BiocSingular	1.26.1	openssl	2.4.1
BiocVersion	3.22.0	org.At.tair.db	3.22.0
Biostrings	2.78.0	otel	0.2.0
bit	4.6.0	pak	0.9.5
bit64	4.8.2	parallelly	1.47.0
bitops	1.0-9	patchwork	1.3.2
blob	1.3.0	pbapply	1.7-4
boot	1.3-32	pbkrtest	0.5.5
BPCells	0.3.1	pbcaply	1.5.1
brew	1.0-10	pheatmap	1.0.13
brio	1.1.5	pillar	1.11.1
broom	1.0.13	pkgbuild	1.4.8
bslib	0.11.0	pkgconfig	2.0.3
cachem	1.1.0	pkgdown	2.2.0
Cairo	1.7-0	pkgload	1.5.2
callr	3.7.6	plotly	4.12.0
car	3.1-5	plyr	1.8.9
carData	3.0-6	png	0.1-9
caTools	1.18.3	polyclip	1.10-7
cellranger	1.1.0	polynom	1.4-1
checkmate	2.3.4	praise	1.0.0
circlize	0.4.18	preprocessCore	1.72.0
class	7.3-23	prettyunits	1.2.0
classInt	0.4-11	processx	3.9.0
cli	3.6.6	profvis	0.4.0
clipr	0.8.1	progress	1.2.3
clue	0.3-68	progressr	0.19.0
cluster	2.1.8.2	promises	1.5.0
clusterProfiler	4.18.4	proxy	0.4-29
clustree	0.5.1	ps	1.9.3
codetools	0.2-20	pscl	1.5.9
colorspace	2.1-2	purrr	1.2.2
commonmark	2.0.0	quadprog	1.5-8
ComplexHeatmap	2.26.1	quantreg	6.1
conflicted	1.2.0	qvalue	2.42.0
corrplot	0.95	R.methodsS3	1.8.2
cowplot	1.2.0	R.oo	1.27.1
cpp11	0.5.5	R.utils	2.13.0

Package	Version	Package	Version
crayon	1.5.3	R6	2.6.1
credentials	2.0.3	ragg	1.5.2
crosstalk	1.2.2	RANN	2.6.2
curl	7.1.0	rappdirs	0.3.4
data.table	1.18.4	rbibutils	2.4.1
DBI	1.3.0	rcmdcheck	1.4.0
dbplyr	2.5.2	RColorBrewer	1.1-3
DelayedArray	0.36.1	Rcpp	1.1.1-1.1
DelayedMatrixStats	1.32.0	RcppAnnoy	0.0.23
deldir	2.0-4	RcppArmadillo	15.2.7-1
Deriv	4.2.0	RcppEigen	0.3.4.0.2
desc	1.4.3	RcppHNSW	0.7.0
DESeq2	1.50.2	RcppML	0.3.7.1
devtools	2.5.2	RcppProgress	0.4.2
diffobj	0.3.6	RcppRoll	0.3.2
digest	0.6.39	RcppTOML	0.2.3
dir.expiry	1.18.0	Rdpack	2.6.6
distributional	0.7.0	readr	2.2.0
doBy	4.7.1	readxl	1.5.0
docopt	0.7.2	reformulas	0.4.4
doParallel	1.0.17	rematch	2.0.0
DOSE	4.4.0	rematch2	2.1.2
dotCall64	1.2	remotes	2.5.0
DoubletFinder	2.0.6	reprex	2.1.1
downlit	0.4.5	reshape2	1.4.5
dplyr	1.2.1	ResidualMatrix	1.20.0
dqrng	0.4.1	reticulate	1.46.0
dtplyr	1.3.3	rhdf5	2.54.1
dynamicTreeCut	1.63-1	rhdf5filters	1.22.0
e1071	1.7-17	Rhdf5lib	1.32.0
ellipsis	0.3.3	RhpcBLASctl	0.23-42
enrichplot	1.30.5	Rhtslib	3.6.0
enrichR	3.4	rjson	0.2.23
eulerr	8.0.0	rlang	1.2.0
evaluate	1.0.5	rmarkdown	2.31
fansi	1.0.7	ROCR	1.0-12
farver	2.1.2	roxygen2	8.0.0
fastcluster	1.3.0	rpart	4.1.24
fastDummies	1.7.6	rprojroot	2.1.1
fastmap	1.2.0	rsample	1.3.2
fastmatch	1.1-8	Rsamtools	2.26.0
fgsea	1.36.2	RSpectra	0.16-2
fields	17.3	RSQLite	3.53.1
filelock	1.0.3	rstatix	0.7.3
fitdistrplus	1.2-6	rstudioapi	0.18.0
FNN	1.1.4.1	rsvd	1.0.5
fontawesome	0.5.3	Rtsne	0.17
fontBitstreamVera	0.1.1	rversions	3.0.0
fontLiberation	0.1.0	rvest	1.0.5
fontquiver	0.2.1	s2	1.1.11
forcats	1.0.1	S4Arrays	1.10.1
foreach	1.5.2	S4Vectors	0.48.1
forecast	9.0.2	S7	0.2.2
foreign	0.8-91	sass	0.4.10
formatR	1.14	ScaledMatrix	1.18.0
Formula	1.2-5	scales	1.4.0
fracdiff	1.5-4	scater	1.38.1
fs	2.1.0	scattermore	1.2
furrr	0.4.0	scatterpie	0.2.6
futile.logger	1.4.9	sctransform	0.4.3
futile.options	1.0.1	scuttle	1.20.0
future	1.70.0	selectr	0.5-1
future.apply	1.20.2	Seqinfo	1.0.0
gargle	1.6.1	sessioninfo	1.2.3
gdtools	0.5.1	Seurat	5.5.0
GeneOverlap	1.46.0	SeuratDisk	0.0.0.9021

Package	Version	Package	Version
generics	0.1.4	SeuratObject	5.4.0
GENIE3	1.32.0	SeuratWrappers	0.4.0
GenomeInfoDb	1.46.2	sf	1.1-1
GenomicRanges	1.62.1	shape	1.4.6.1
gert	2.3.1	shiny	1.13.0
GetoptLong	1.1.1	SingleCellExperiment	1.32.0
ggbeeswarm	0.7.3	sitmo	2.0.2
ggdist	3.3.3	slam	0.1-55
ggforce	0.5.0	slider	0.3.3
ggfun	0.2.0	snow	0.4-4
ggiraph	0.9.6	sourcetools	0.1.7-2
ggnewscale	0.5.2	sp	2.2-1
ggplot2	4.0.3	spam	2.11-4
ggplotify	0.1.3	SparseArray	1.10.10
ggpubr	0.6.3	SparseM	1.84-2
ggraph	2.2.2	sparseMatrixStats	1.22.0
ggrastr	1.0.2	spatial	7.3-18
ggrepel	0.9.8	spatstat.data	3.1-9
ggridges	0.5.7	spatstat.explore	3.8-1
ggsci	5.0.0	spatstat.geom	3.8-1
ggsignif	0.6.4	spatstat.random	3.5-0
ggtangle	0.1.2	spatstat.sparse	3.2-0
ggtree	4.0.5	spatstat.univar	3.2-0
ggvenn	0.1.19	spatstat.utils	3.2-3
gh	1.6.0	spData	2.3.5
gitcreds	0.1.2	spdep	1.4-2
GlobalOptions	0.1.4	speedglm	0.3-4
globals	0.19.1	statmod	1.5.2
glue	1.8.1	stringi	1.8.7
G0.db	3.22.0	stringr	1.6.0
goftest	1.2-3	SummarizedExperiment	1.40.0
googledrive	2.1.2	survival	3.8-6
googlesheets4	1.1.2	svglite	2.2.2
GOSemSim	2.36.0	sys	3.4.3
gplots	3.3.0	systemfonts	1.3.2
graphlayouts	1.2.3	tensor	1.5.1
gridExtra	2.3	tester	0.3.0
gridGraphics	0.5-1	testthat	3.3.2
grr	0.9.5	textshaping	1.0.5
gson	0.1.0	tibble	3.3.1
gtable	0.3.6	tidydr	0.0.6
gtools	3.9.5	tidygraph	1.3.1
h5mread	1.2.1	tidyr	1.3.2
harmony	2.0.3	tidyselect	1.2.1
haven	2.5.5	tidytree	0.4.7
HDF5Array	1.38.0	tidyverse	2.0.0
hdf5r	1.3.12	timechange	0.4.0
hdWGCNA	0.4.11	timeDate	4052.112
here	1.0.2	tinytex	0.59
hexbin	1.28.5	treeio	1.34.0
highr	0.12	tweenr	2.0.3
Hmisc	5.2-5	tzdb	0.5.0
hms	1.1.4	UCCell	2.14.0
htmlTable	2.5.0	UCSC.utils	1.6.1
htmltools	0.5.9	units	1.0-1
htmlwidgets	1.6.4	UpSetR	1.4.1
httpuv	1.6.17	urca	1.3-4
httr	1.4.8	urlchecker	1.0.1
httr2	1.2.2	usethis	3.2.1
ica	1.0-3	utf8	1.2.6
ids	1.0.1	uuid	1.2-2
igraph	2.3.2	uwot	0.2.4
impute	1.84.0	vctrs	0.7.3
ini	0.3.1	VennDiagram	1.8.2
IRanges	2.44.0	vipor	0.4.7
irlba	2.3.7	viridis	0.6.5

Package	Version	Package	Version
isoband	0.3.0	viridisLite	0.4.3
iterators	1.0.14	vroom	1.7.1
jquerylib	0.1.4	waldo	0.6.2
jsonlite	2.0.0	warp	0.2.3
kableExtra	1.4.0	WGCNA	1.74
KEGGREST	1.50.0	whisker	0.4.1
KernSmooth	2.23-26	withr	3.0.2
knitr	1.51	wk	0.9.5
labeling	0.4.3	WriteXLS	6.8.0
lambda.r	1.2.4	xfun	0.58
later	1.4.8	xml2	1.5.2
lattice	0.22-9	xopen	1.0.1
lazyeval	0.2.3	xtable	1.8-8
leidenbase	0.1.37	XVector	0.50.0
lifecycle	1.0.5	yaml	2.3.12
limma	3.66.0	yulab.utils	0.2.4
listenv	0.10.1	zellkonverter	1.20.1
littler	0.3.23	zip	2.3.3
lme4	2.0-1	zoo	1.8-15
lmtest	0.9-40		

Python packages (207)

Package	Version	Package	Version
adjustText	1.3.0	notebook	6.5.7
anndata	0.12.16	notebook_shim	0.2.4
annotated-types	0.7.0	numba	0.65.1
anyio	4.13.0	numcodecs	0.16.5
argon2-cffi	25.1.0	numexpr	2.14.1
argon2-cffi-bindings	25.1.0	numpy	2.4.6
array-api-compat	1.14.0	numpy-groupies	0.11.3
arrow	1.4.0	nvidia-cublas	13.1.1.3
asttokens	3.0.1	nvidia-cuda-cupti	13.0.85
async-lru	2.3.0	nvidia-cuda-nvrtc	13.0.88
attrs	26.1.0	nvidia-cuda-runtime	13.0.96
babel	2.18.0	nvidia-cudnn-cu13	9.20.0.48
beautifulsoup4	4.14.3	nvidia-cufft	12.0.0.61
bleach	6.3.0	nvidia-cufile	1.15.1.6
blosc2	4.4.1	nvidia-curand	10.4.0.35
cellbender	0.3.2	nvidia-cusolver	12.0.4.66
certifi	2026.5.20	nvidia-cusparse	12.6.3.3
cffi	2.0.0	nvidia-cusparselt-cu13	0.8.1
charset-normalizer	3.4.7	nvidia-nccl-cu13	2.29.7
click	8.4.1	nvidia-nvjitlink	13.0.88
comm	0.2.3	nvidia-nvshmem-cu13	3.4.5
contourpy	1.3.3	nvidia-nvtx	13.0.85
cuda-bindings	13.3.1	opt_einsum	3.4.0
cuda-pathfinder	1.5.5	packaging	26.2
cuda-toolkit	13.0.2	palantir	1.4.4
cycler	0.12.1	pandas	2.3.3
debugpy	1.8.21	pandocfilters	1.5.1
decorator	5.3.1	parso	0.8.7
defusedxml	0.7.1	patsy	1.0.2
donfig	0.8.1.post1	pexpect	4.9.0
elpigraph-python	0.3.4	pillow	12.2.0
entrypoints	0.4	pip	26.1.2
executing	2.2.1	platformdirs	4.10.0
fa2_modified	0.4	plotly	6.7.0
fast-array-utils	1.4.1	prometheus_client	0.25.0
fastjsonschema	2.21.2	prompt_toolkit	3.0.52
filelock	3.29.0	psutil	7.2.2
fonttools	4.63.0	ptyprocess	0.7.0
fqdn	1.5.1	pure_eval	0.2.3

Package	Version	Package	Version
fsspec	2026.4.0	py-cpuinfo	9.0.0
google-crc32c	1.8.0	pycparser	3.0
h11	0.16.0	pydantic	2.13.4
h5py	3.16.0	pydantic_core	2.46.4
httpcore	1.0.9	Pygments	2.20.0
httpx	0.28.1	pynndescent	0.6.0
idna	3.18	pyparsing	3.3.2
igraph	1.0.0	pyro-api	0.1.2
ipykernel	6.29.5	pyro-ppl	1.9.1
ipython	9.14.0	python-dateutil	2.9.0.post0
ipython-genutils	0.2.0	python-igraph	1.0.0
ipython-pygments_lexers	1.1.1	python-json-logger	4.1.0
ipywidgets	8.1.8	pytz	2026.2
isoduration	20.11.0	PyYAML	6.0.3
jax	0.10.1	pyzmq	27.1.0
jaxlib	0.10.1	referencing	0.37.0
jaxopt	0.8.5	requests	2.34.2
jedi	0.20.0	rfc3339-validator	0.1.4
Jinja2	3.1.6	rfc3986-validator	0.1.1
joblib	1.5.3	rfc3987-syntax	1.1.0
json5	0.14.0	rpds-py	2026.5.1
jsonpointer	3.1.1	rpy2	3.6.7
jsonschema	4.26.0	rpy2-rinterface	3.6.6
jsonschema-specifications	2025.9.1	rpy2-robjects	3.6.5
jupyter	1.1.1	scanpy	1.12.1
jupyter-console	6.6.3	scFates	1.2.4
jupyter-events	0.12.1	scikit-learn	1.9.0
jupyter-highlight-selected-word	0.2.0	scikit-misc	0.5.2
jupyter-lsp	2.3.1	scipy	1.17.1
jupyter_client	7.4.9	scverse-misc	0.0.7
jupyter_contrib_core	0.4.2	seaborn	0.13.2
jupyter_contrib_nbextensions	0.7.0	Send2Trash	2.1.0
jupyter_core	5.9.1	session-info2	0.4.1
jupyter_nbextensions_configurator	0.6.4	setuptools	81.0.0
jupyter_server	2.19.0	shapely	2.1.2
jupyter_server_terminals	0.5.4	simpleppt	1.1.7
jupyterlab	4.5.7	six	1.17.0
jupyterlab_pygments	0.3.0	soupsieve	2.8.4
jupyterlab_server	2.28.0	stack-data	0.6.3
jupyterlab_widgets	3.0.16	statsmodels	0.14.6
kiwisolver	1.5.0	sympy	1.14.0
lark	1.3.1	tables	3.11.1
legacy-api-wrap	1.5	terminado	0.18.1
leidenalg	0.12.0	texttable	1.7.0
llvmlite	0.47.0	threadpoolctl	3.6.0
loompy	3.0.8	tinycss2	1.5.1
lxml	6.1.1	torch	2.12.0
lxml_html_clean	0.4.5	tornado	6.5.6
MarkupSafe	3.0.3	tqdm	4.67.3
matplotlib	3.10.9	traitlets	5.15.0
matplotlib-inline	0.2.2	triton	3.7.0
mellon	1.7.1	typing-inspection	0.4.2
mistune	0.8.4	typing_extensions	4.15.0
ml_dtypes	0.5.4	tzdata	2026.2
mpmath	1.3.0	tzlocal	5.4.4
msgpack	1.1.2	umap-learn	0.5.12
narwhals	2.22.0	uri-template	1.3.0
natsort	8.4.0	urllib3	2.7.0
nbclassic	1.3.3	wcwidth	0.7.0
nbclient	0.10.4	webcolors	25.10.0
nbconvert	6.5.4	webencodings	0.5.1
nbformat	5.10.4	websocket-client	1.9.0
ndindex	1.10.1	widetsnbextension	4.0.15
nest-asyncio	1.6.0	zarr	3.2.1
networkx	3.6.1		